Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 March 2026
Team ID	69c0f3744241397902eb6826
Project Name	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
Maximum Marks	4 Marks

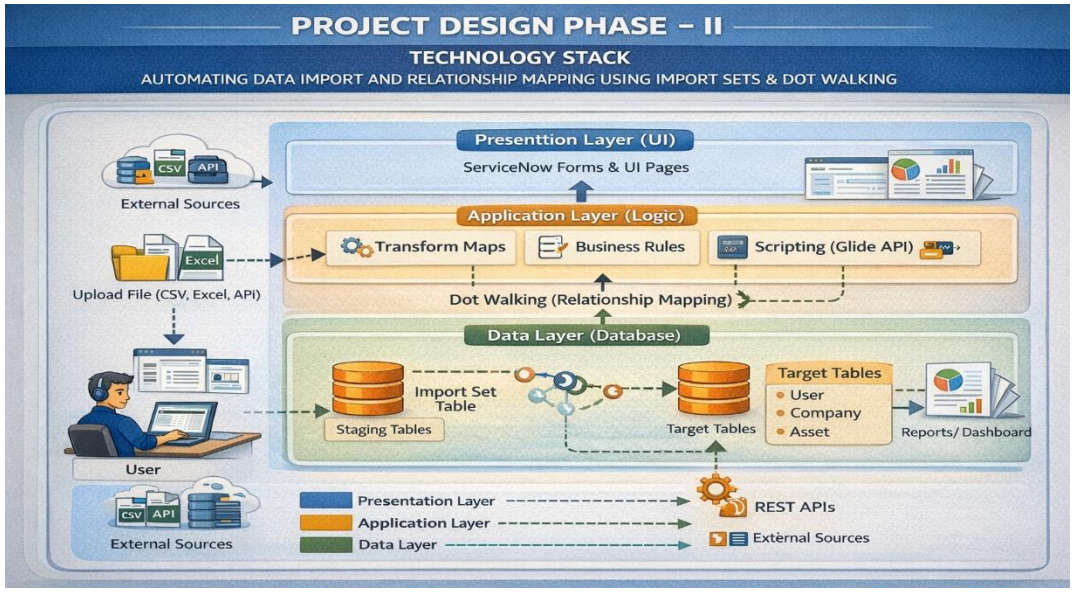
◈ Technology Stack (Architecture & Stack)

(Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking)

🏗 System Architecture (Overview)

The system follows a **layered architecture**:

1. **Presentation Layer (UI)**
   - ServiceNow Forms & UI Pages
   - Used for uploading data and viewing reports
2. **Application Layer (Logic)**
   - Transform Maps
   - Business Rules



- Scripts (Glide API)
- Handles data processing and mapping

3. **Data Layer (Database)**
    - Import Set Tables (staging tables)
    - Target Tables (User, Company, Asset)
    - Stores structured data

🔧 **Technology Stack (Table)**

LAYER	TECHNOLOGY / TOOL	DESCRIPTION
FRONTEND	ServiceNow UI	Forms and dashboards for user interaction
BACKEND	ServiceNow Platform	Core platform handling logic and workflows
SCRIPTING	JavaScript (Glide API)	Used for Transform Scripts & Business Rules
DATA IMPORT	Import Sets	Used to load external data into staging tables
DATA MAPPING	Transform Maps	Maps source fields to target tables
RELATIONSHIP MAPPING	Dot Walking	Links related tables dynamically
DATABASE	ServiceNow Tables	Stores data (Import Set & Target Tables)
INTEGRATION	REST APIs	Used for external system integration

FILE HANDLING	CSV / Excel	Common formats for data import
REPORTING	ServiceNow Reports	Generates visual reports and dashboards

🔄 Architecture Flow (Simple)

User → Upload File (CSV/Excel/API)
↓
Import Set Table (Staging)
↓
Transform Map (Processing)
↓
Dot Walking (Relationship Mapping)
↓
Target Tables (Database)
↓
Reports / Dashboard

🎯 Key Highlights

- **ServiceNow Platform** is the core technology
- **Import Sets + Transform Maps** handle automation
- **Dot Walking** ensures relationship mapping

◈ Components & Technologies

S.No	Component	Technology Used	Description
1	User Interface	ServiceNow UI	Allows users to upload files and view reports
2	Data Import Module	Import Sets	Handles importing external data into staging tables
3	Data Transformation	Transform Maps	Maps and converts source data to target tables
4	Relationship Mapping	Dot Walking	Links related records across tables
5	Business Logic	Business Rules	Automates backend processing and validation
6	Scripting Engine	JavaScript (Glide API)	Custom scripts for transformation and logic
7	Database	ServiceNow Tables	Stores Import Set and target data
8	Integration Layer	REST APIs	Connects external systems to ServiceNow
9	File Handling	CSV / Excel	Supports structured data import formats
10	Reporting Module	ServiceNow Reports	Generates dashboards and reports

⚙ Application Characteristics

S.No	Characteristic	Description
1	Automation	Reduces manual effort in data import and mapping
2	Accuracy	Ensures correct data mapping using Transform Maps
3	Efficiency	Speeds up data processing and reduces time
4	Scalability	Handles large volumes of data and users
5	Reliability	Provides consistent and error-free operations
6	Flexibility	Supports multiple data formats (CSV, Excel, API)
7	Security	Ensures authorized access and data protection
8	Usability	Easy-to-use interface for non-technical users
9	Maintainability	Easy to update mappings and configurations
10	Integration Capability	Seamlessly connects with external systems